CS Capstone Design

Product Demonstration Grading Sheet (100 pts)

TEAM: KIMinder360 Team

PROJECT: KIMinder360 Mobile App

DEVELOPMENT CYCLE: ☑ Alpha Demo I ☐ Alpha Demo II ☐Acceptance Demo

DATE: 2/6/2026

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CLIENT: Morgan Boatman

1. Overview

The purpose of the Product Demonstration Flight Plan is to define what functionality is planned, what functionality will be demonstrated in the current development cycle, and how success will be evaluated.

This document serves three roles:

1. A **planning artifact**, showing how the full product scope is distributed across development cycles
2. A **demonstration guide**, outlining how the current cycle's functionality will be presented through realistic user flows
3. An **evaluation aid**, defining acceptance criteria used by the mentor and client to assess completeness and correctness

The Flight Plan is prepared by the team, reviewed and approved by the mentor, and used during the live demonstration for evaluation and feedback.

2. Product Scope and Development Cycles Overview

This section presents the full set of user stories or use cases for the product and how they are distributed across the three development cycles. The intent is to make clear:

- What functionality exists in the overall product, corresponding to your full prioritized list of functional requirements
- What is included in each development cycle
- How incremental delivery builds toward the final system

2.1. Backlog Overview by Development Cycle

Alpha Demo I – Current Cycle

ID	User Story / Use Case	Acceptance Criteria (Summary)
US-1	As a user, I want to select active hours so that I can receive directives when I'm able to complete them	Active hours configurable per day, saved, active/inactive status shown on home page based on current time
US-2	As a user, I want to select different categories of directives to keep my training relevant and engaging	Categories selectable, saved, only selected categories marked active
US-3	As a user, I want to receive directives through push notifications so that I can practice situational awareness throughout my day	Notifications sent only during active hours, directives limited to selected categories
US-4	As a user, I want directives to be randomized so training remains fresh and engaging	Notification sent with directive, chosen randomly from active categories

US-5	As a user, I want directives to be sent at random times within my active hours so training is spontaneous unpredictable	Notification sent with directive, sent at random point within active hours
US-6	As a user, I want to see a list of my most recent directives so I can track my training progress	Recent directives tracked in database, visible on home screen with date sent
US-7	As a user, I want to be able to mark directives as complete so I can keep myself accountable	Directives can be marked complete or incomplete, interaction level based on settings, response saved and shown in recent directives list
US-8	As someone with a schedule that fluctuates, I want to be able to change how much I need to interact with directives depending on if I have a busy day or a relaxed day	Interaction level selectable, response requirements enforced based on selection
US-9	As someone with different free time throughout the week, I want to be able to increase or decrease directive frequency based on my schedule	Directive sending frequency configurable, settings measurably affect real frequency of directive sending

Alpha Demo II - Planned Scope

ID	User Story / Use Case	Acceptance Criteria (Summary)
US-10	As a user, I want to create my own directives so that I can personalize my experience and focus on training goals that are meaningful to me	User can create directives and categories, stored locally on device, included in notification selection
US-11	As an admin, I want to be able to add and update directives and categories for future app expansion	Directives and categories addable by admin users, included in app updates, accessible to users after update
US-12	As a user, I want to be able to skip, delay, or bookmark a directive so that I can manage my tasks according to my schedule and complete them at a more convenient time	Directive can be skipped or delayed, delayed directives are resent at later selected time

Acceptance Demo - Planned Scope

ID	User Story / Use Case	Acceptance Criteria (Summary)
US-12	As a user, I want to be able to change the difficulty level of my directives so that I can have an experience that lets me grow my skills and challenge myself	Difficulty level selectable, directives filtered by selected difficulty
US-14	As someone who travels a lot, I want directives to adapt to my current location so that I can connect with other users around me and explore the new locations I visit	Specific directives triggered by location, sent only when location conditions are met
US-15	As someone who plays a lot of games, I want directives that span multiple parts or steps so that I can engage in progressive challenges similar to quests in games	Directives contain multiple steps, progress tracked, completion requires all parts
US-16	As a user, I want to view and compare my directive completion progress on a leaderboard so that I can be motivated by others to complete my directives	Leaderboard displays user progress rankings, updates based on completed directives

3. Demonstration Walkthrough

For each user story or use case included in the current cycle, provide a demonstration walkthrough guided by acceptance criteria. The demo should follow a realistic user flow and show the functionality working end-to-end.

Use Case / User Story: US-1 - As a user, I want to select active hours so that I can receive directives when I'm able to complete them

User Role: Regular user training situational awareness

Goal: Ensure directives are only delivered at times when the user is available and prepared to engage with them.

Demonstration flow:

1. User starts at home screen
2. User opens active hour selection screen
3. User configures active hours for each day of the week
4. User saves hour selections
5. System records active hours in database
6. Home screen shows whether notifications are currently active or not

Acceptance Criteria (What Success Looks Like)

The demonstration is considered successful if:

1. Active hours screen opens from home and shows 7 days with current settings
2. User can set and start end times for each day and see changes immediately
3. System stores active hours upon tapping "Save" and can be reopened
4. Home screen correctly displays active vs. inactive based on saved hours
5. Directives only send during active hours and do not send during inactive hours

Evaluation and Comments (for mentors only):

- Convincingly demonstrated the functionality

- Other evaluative comments/notes

Use Case / User Story: US-2 - As a user, I want to select different categories of directives to keep my training relevant and engaging

User Role: User tailoring their training to their environment or interests

Goal: Receive only relevant directive types so training feels more intentional

Demonstration flow:

1. User starts at the home screen of the KIMinder360 app.
2. User selects the "Categories" button to open the category selection screen
3. User unchecks multiple categories
4. User selects "Cancel" to discard changes
5. System restores previously saved category selections
6. User reopens categories selection screen

7. User deselects all categories and then selects four desired categories
8. User selects “Save” to confirm the changes
9. System saves the selected categories
10. User reopens the “Categories” selection screen to verify that saved selections are preserved

Acceptance Criteria (What Success Looks Like)

The demonstration is considered successful if:

1. Category screen displays the current saved selections
2. Tapping “Cancel” restores previous selections without saving changes
3. “Save” stores updated category selections
4. Saved selections remain consistent when the screen is reopened

Evaluation and Comments (for mentors/clients only):

- Convincingly demonstrated the functionality

- Other evaluative comments/notes

Use Case / User Story: US-3 - As a user, I want to receive directives through push notifications so that I can practice situational awareness throughout my day

User Role: Active user integrating situational awareness practice into daily life

Goal: Be prompted through push notifications to practice situational awareness throughout the day without ever needing to open the app manually

Demonstration flow:

1. User starts at home screen with app closed
2. User performs a swipe-down gesture to open the notification shade
3. User taps on KIMinder360 notifications group to view notifications
4. User taps on a specific notification
5. System opens the KIMinder360 app and displays the content

Acceptance Criteria (What Success Looks Like)

The demonstration is considered successful if:

1. KIMinder360 notification appears while the app is closed
2. Notification is correctly grouped in KIMinder360
3. Tapping the notification opens the app
4. System displays correct directive content

Evaluation and Comments (for mentors/clients only):

- Convincingly demonstrated the functionality

- Other evaluative comments/notes

Use Case / User Story: US-4 - As a user, I want directives to be randomized so training remains fresh and engaging

User Role: User regularly performing situational awareness training

Goal: Experience varied directive prompts without a fixed order or predictable sequence

Demonstration flow:

1. User has already selected at least one directive category to be active
2. System selects a directive at random from the pool of active categories
3. System sends that directive as a notification during active hours
4. User receives additional directives over time
5. User navigates to the home screen to view the recent directives list
6. User observes that directives are selected randomly
7. User confirms all received directives belong to active categories

Acceptance Criteria (What Success Looks Like)

The demonstration is considered successful if:

1. Each directive is selected randomly from active categories
2. No predefined order or sequence is used when selecting directives
3. Repeat directives may occur due to fully random selection
4. All directives shown in the recent directives list come from active categories

Evaluation and Comments (for mentors/clients only):

- Convincingly demonstrated the functionality

- Other evaluative comments/notes

Use Case / User Story: US-5 - As a user, I want directives to be sent at random times within my active hours so training is spontaneous unpredictable

User Role: User practicing situational awareness training

Goal: Train awareness in realistic, unscheduled moments

Demonstration flow:

1. User starts on home screen
2. User has configured active hours for the current day and at least one active directive category
3. System waits a randomized amount of time within that window
4. A directive notification is delivered
5. User notes that the timing was not fixed or scheduled
6. User receives additional directives later at different, randomized times

Acceptance Criteria (What Success Looks Like)

The demonstration is considered successful if:

1. Directives are sent only within active hours
2. Notification timing varies day-to-day and within each active window
3. No predictable interval pattern is observed
4. No notifications are sent outside active hours

Evaluation and Comments (for mentors/clients only):

- Convincingly demonstrated the functionality

- Other evaluative comments/notes

Use Case / User Story: US-6 - As a user, I want to see a list of my most recent directives so I can track my training progress

User Role: User reviewing past training activity

Goal: Reflect on consistency and participation over time

Demonstration flow:

1. User starts on home screen
2. User views the recent directives list
3. System displays directives sorted by most recently sent first
4. Each directive shows its content and date sent
5. User taps a directive to view interaction details

Acceptance Criteria (What Success Looks Like)

The demonstration is considered successful if:

1. Recent directives appear automatically on home screen
2. Each entry includes directive text and timestamp
3. List updates as new directives are sent

Evaluation and Comments (for mentors/clients only):

- Convincingly demonstrated the functionality

- Other evaluative comments/notes

Use Case / User Story: US-7 - As a user, I want to be able to mark directives as complete so I can keep myself accountable

User Role: User actively completing directives

Goal: Acknowledge completion and reinforce training habits

Demonstration flow:

1. User starts on home screen
2. User opens a directive from the recent directives list
3. System displays interaction UI based on set interaction level
4. User marks directive as complete
    a. Marks checkbox, writes in free-text response box describing observations, if enabled
5. User saves interaction
6. System records updated directive status

7. User returns to home screen

Acceptance Criteria (What Success Looks Like)

The demonstration is considered successful if:

1. Completion UI matches selected interaction level
2. Completion state is saved in database

Evaluation and Comments (for mentors/clients only):

- Convincingly demonstrated the functionality

- Other evaluative comments/notes

Use Case / User Story: US-8 - As someone with a schedule that fluctuates, I want to be able to change how much I need to interact with directives depending on if I have a busy day or a relaxed day

User Role: User adjusting training based on daily availability

Goal: User can interact with directives different amounts according to their personal preference

Demonstration flow:

1. User starts on home screen
2. User opens settings screen
3. User adjusts interaction level slider
4. User saves settings
5. System updates interaction requirements
6. User receives a new directive
7. User opens directive from recent directives list and sees updated interaction UI

Acceptance Criteria (What Success Looks Like)

The demonstration is considered successful if:

1. Interaction levels can be changed at any time
2. Settings persist after saving
3. New directives enforce the selected interaction level
4. UI adjusts correctly for each interaction level

Evaluation and Comments (for mentors/clients only):

- Convincingly demonstrated the functionality

- Other evaluative comments/notes

Use Case / User Story: US-9 - As someone with different free time throughout the week, I want to be able to increase or decrease directive frequency based on my schedule

User Role: User managing training intensity

Goal: Control how often directives interrupt daily routines

Demonstration flow:

1. User starts at home screen
2. User opens settings screen
3. User adjusts directive frequency slider
4. User saves settings
5. System reruns notification scheduling process
6. User observed altered directive delivery frequency over time

Acceptance Criteria (What Success Looks Like)

The demonstration is considered successful if:

1. Frequency slider changes are saved correctly
2. Higher settings result in more frequent directives
3. Lower settings result in less frequent directives
4. Changes are reflected in notification behavior

Evaluation and Comments (for mentors/clients only):

- Convincingly demonstrated the functionality

- Other evaluative comments/notes

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4. Out-of-Scope or Deferred Functionality

List any user stories or use cases planned in the approved flight plan for this cycle, but not demonstrated, along with their planned future cycle. This helps set clear expectations and avoids confusion during the demo.

Alpha Demo I – Current Cycle

ID	Reason Not Demonstrated	Planned Cycle
N/A		

5. Known Limitations and Planned Follow-Up

Document any known limitations (if any), partial implementations, or issues observed during development or demonstration. Where possible, indicate planned mitigation or timing for resolution.

- Limitation / Issue: Recent directives will stay in local storage forever leading to increased storage usage
- Impact: After using the app for an extended period of time, there will be increased storage usage
- Planned Resolution (if known): Add user controllable deletion settings for recent directives

6. Approval

This Flight Plan has been reviewed and approved for use in the Alpha Demonstration.

Mentor Approval: [can be digitally signed]

Date: [can be digitally signed]